

SUOLOVUOPMI LULIT, Norway

WMO: 010580

Lat: 69.579N

Lon: 23.534E

Elev: 1250

StdP: 14.04

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.2	-22.2	-33.4	0.9	-26.8	-27.8	1.2	-21.6	29.4	21.4	26.6	21.8	3.8	130	0.820

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	70.9	56.7	66.8	55.4	63.0	53.8	59.3	66.7	56.8	63.6	54.9	61.5	8.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.9	69.8	62.3	53.6	63.9	59.2	51.5	59.2	57.8	26.5	66.6	24.8	63.7	23.6	61.6	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.8	20.2	17.6	-32.8	77.7	5.1	5.3	-36.5	81.5	-39.5	84.6	-42.4	87.6	-46.1	91.5		
			-32.6	61.9	5.2	3.2	-36.3	64.2	-39.3	66.0	-42.2	67.8	-46.0	70.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	7.3	8.3	14.0	24.2	36.3	45.8	52.8	49.6	42.3	29.2	17.7	14.1
	DBStd	18.56	13.67	14.05	11.95	8.94	7.29	6.71	6.50	6.01	6.06	8.61	10.38	11.95
	HDD50	8061	1323	1167	1116	773	429	163	43	83	238	645	970	1111
	HDD65	13289	1788	1587	1581	1223	889	575	381	479	680	1110	1420	1576
	CDD50	251	0	0	0	0	5	38	130	70	8	0	0	0
	CDD65	4	0	0	0	0	0	0	3	1	0	0	0	0
	CDH74	70	0	0	0	0	4	2	51	13	0	0	0	0
	CDH80	10	0	0	0	0	0	0	8	2	0	0	0	0
Wind	WSAvg	7.9	8.0	8.6	8.9	8.2	8.2	8.3	7.6	6.8	7.7	7.0	7.2	8.0
Precipitation	PrecAvg	18.0	1.2	1.0	1.1	0.9	1.2	2.0	2.6	2.3	1.5	1.7	1.3	1.3
	PrecMax	21.5	2.4	1.7	2.0	1.9	2.4	3.8	5.8	4.7	2.6	3.9	2.4	2.8
	PrecMin	12.4	0.4	0.1	0.5	0.2	0.2	0.6	0.7	1.1	0.3	0.5	0.3	0.6
	PrecStd	2.2	0.5	0.4	0.3	0.4	0.6	0.9	1.3	0.9	0.6	0.8	0.6	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.0	37.9	39.0	46.1	68.1	70.9	80.4	74.2	61.8	48.8	37.8	38.8
		MCWB	33.3	35.1	33.7	39.6	52.1	54.0	63.3	60.4	52.1	44.5	35.8	35.6
	2%	DB	31.6	33.5	34.8	41.4	59.1	66.2	72.9	68.2	57.6	45.4	35.7	35.3
		MCWB	29.9	30.8	31.6	35.7	46.6	53.5	58.7	56.8	50.7	42.4	34.1	32.7
	5%	DB	28.3	30.1	32.2	38.7	52.0	61.4	68.6	63.6	54.5	42.4	33.1	32.0
		MCWB	26.8	28.5	29.6	34.2	43.4	51.0	56.6	54.8	49.2	39.9	31.7	30.4
	10%	DB	25.2	26.9	29.8	36.4	47.3	57.6	64.7	60.0	51.9	39.8	30.6	29.7
		MCWB	24.0	25.4	27.7	32.6	40.6	49.9	55.3	53.7	47.5	37.9	29.5	28.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.7	35.1	34.2	40.2	53.0	57.4	65.0	62.9	54.7	44.5	36.4	36.0
		MCDB	35.7	37.7	38.0	46.2	68.4	64.2	77.3	73.6	58.9	47.5	37.4	38.7
	2%	WB	29.9	31.0	32.1	36.4	47.5	55.4	60.9	58.0	52.0	42.3	34.2	33.0
		MCDB	31.6	33.1	34.8	40.6	58.1	61.6	69.6	63.5	56.1	45.0	35.5	34.9
	5%	WB	26.9	28.5	29.8	34.5	43.7	52.9	58.1	55.7	49.9	40.1	31.8	30.4
		MCDB	28.3	30.1	32.0	38.4	51.1	60.8	65.9	61.7	53.8	42.1	32.9	31.9
	10%	WB	24.1	25.5	27.9	32.9	41.2	50.0	55.9	54.0	47.7	38.0	29.7	28.3
		MCDB	25.1	26.7	29.9	36.0	47.0	56.3	63.2	59.1	51.3	39.7	30.7	29.5
Mean Daily Temperature Range	5% DB	MDBR	16.4	16.3	19.0	18.0	13.9	13.7	15.3	15.1	12.1	9.9	13.6	15.5
		MCDBR	16.1	16.3	14.0	15.8	22.6	20.2	22.7	22.1	16.6	10.6	10.9	13.3
		MCWBR	15.2	14.9	11.8	11.3	12.6	10.1	10.2	12.1	10.2	8.2	10.2	12.1
	5% WB	MCDBR	15.9	15.7	13.8	15.2	21.5	18.2	20.0	18.4	14.4	9.7	10.5	13.0
		MCWBR	15.1	14.4	12.0	11.0	12.4	9.7	10.1	10.9	9.5	7.9	10.0	12.0
Clear-Sky Solar Irradiance	taub	0.181	0.229	0.277	0.301	0.319	0.323	0.330	0.316	0.288	0.241	N/A	N/A	
	taud	1.911	2.068	2.171	2.223	2.361	2.500	2.505	2.524	2.558	2.227	N/A	N/A	
	Ebn at Noon	9	164	233	265	275	278	271	262	241	160	0	0	
	Edh at Noon	1	13	24	31	31	28	27	23	17	11	0	0	
All-Sky Solar Radiation	RadAvg	4	117	567	1200	1597	1547	1419	989	517	156	0	0	
	RadStd	0	10	27	57	141	126	137	80	44	14	0	0	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (29 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page